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3. Catalytic Amination: (GS/GOGAT Pathway)

Glutamate amino acid combines with ammonia to form the amide glutamine.

$$\text{Glutamate} + NH_4^+ \xrightarrow[\text{ATP}\quad\text{ADP + Pi}]{\text{Glutamine Synthetase (GS)}} \text{Glutamine.}$$

Glutamine reacts with α ketoglutaric acid to form two molecules of glutamate.

$$\text{Glutamine} + \underset{\text{(2- oxoglutarate)}}{\text{α Ketoglutaric acid}} \xrightarrow[\text{NADH+H}^+\quad\text{NAD}^+]{\text{GOGAT (enzyme)}} \text{2 Glutamate}$$

(GOGAT- Glutamine-2-Oxoglutarate aminotransferase)

12.9 Special modes of nutrition

Nutrition is the process of uptake and utilization of nutrients by living organisms. There are two main types such as **autotrophic** and **heterotrophic** nutrition. Autotrophic nutrition is further divided into **photosynthetic** and **chemosynthetic** nutrition. Heterotrophic nutrition is further divided into saprophytic, parasitic, symbiotic and insectivorous type. In this topic you are going to learn about special mode of nutrition.

12.9.1 Saprophytic mode of nutrition in angiosperms

Saprophytes derive nutrients from dead and decaying matter. Bacteria and fungus are main saprophytic organisms. Some angiosperms also follow saprophytic mode of nutrition. Example: *Neottia*. Roots of *Neottia* (Bird's Nest Orchid) associate with mycorrhizae and absorb nutrients as a saprophyte. *Monotropa* (Indian Pipe) grow on humus rich soil found in thick forests. It absorbs nutrient through mycorrhizal association (Figure 12.9).



Neottia (Bird's Nest Orchid) ***Monotropa*** (Indian Pipe)

Figure 12.9: Saprophytic Mode of nutrition

12.9.2 Parasitic mode of nutrition in angiosperms

Organisms deriving their nutrient from another organism (host) and causing disease to the host are called parasites.

a. Obligate or Total parasite - Completely depends on host for their survival and produces haustoria.

i. **Total stem parasite**: The leafless stem twine around the host and produce haustoria. Example: *Cuscuta* (Dodder), a rootless plant growing on *Zizyphus*, *Citrus* and so on.

ii. **Total root parasite**: They do not have stem axis and grow in the roots of host plants produce haustoria. Example: *Rafflesia*, *Orobanche* and *Balanophora*.

b. Partial parasite - Plants of this group contain chlorophyll and synthesize carbohydrates. Water and mineral requirements are dependent on host plant.

i. **Partial Stem Parasite**: Example: *Loranthus* and *Viscum* (Mistletoe)

Loranthus grows on fig and mango trees and absorb water and minerals from xylem.